

SHIV KRISHNA SINGH

Vill. Kamhariya, Chandauli, Uttar Pradesh

☎ +91-9555960955 ✉ shivkrishnasingh1299@gmail.com 🔗 [LinkedIn](#)

EDUCATION

Indian Institute of Technology (BHU) Varanasi

2022 – 2026

B.Tech - Mechanical Engineering

CGPA: 7.28

SKILLS

Languages & Tech : Python, C++, SQL

Tools & Libraries: TensorFlow, Keras, Scikit-learn, Huggingface, FastAPI, Numpy, Pandas, Optuna

Areas of Interest : Machine Learning, Natural Language Processing (Transformers, Large Language Models, Retrieval Augmented Generation), Data Science, Generative AI, Zero/Few shot Learning, Deep Learning

EXPERIENCE (INTERNSHIP)

NLP Research Intern — IIT(BHU)

May 2025 – July 2025

- Contributed to **low-resource machine translation** by building an English–Sanskrit modern-text corpus.
- Built **preprocessing pipelines** for **175k English–French** and **53k English–Sanskrit** parallel corpora.
- Implemented and fine-tuned seq2seq models (**Helsinki-NLP/opus-mt-en-fr**, **mBART**, **IndicBART**) using **Hugging Face Transformers** with integrated evaluation and performance optimization.
- Achieved BLEU score of 53.72 (English–French) and 27.2 (English–Sanskrit) via optimized inference pipeline

PROJECTS

Graph-RAG based Travel Recommendation [Link](#)

March'25-April'25

- Built Python data pipeline to process and augment location data from **Google Map** and **Wikipedia API**
- Engineered a **Text-to-Cypher** pipeline with **Google's Gemini API** and **prompt engineering** to generate Cypher queries from raw text, directly building the **Neo4j knowledge graph**.
- Implemented **RAG** pipeline querying Neo4j graphs to generate personalized itineraries from user preferences

Customer Offer Prediction | The American Express Campus Challenge [Link](#)

July'25-July'25

- Built a Machine Learning pipeline to predict **top-7 customer offers**, achieving a **MAP@7** score of 0.65.
- Engineered features from transactions, events, and offer metadata, applied **model-specific preprocessing**, and tuned boosting models (**XGBoost**, **LightGBM**, **CatBoost**) with **Optuna** for predictive analytics.
- Designed a **stacking ensemble** with **Logistic Regression** as the **meta-model** to combine base learners, improving ranking performance and recommendation system accuracy.

Agentic AI Contributor for Open-Source Go Projects [Link](#)

Dec'24-Jan'25

- Built a tool-augmented **Agentic AI** system for GitHub issue understanding, fix planning, code generation, validation, and PR creation in open-source Go repositories.
- Developed a repository-aware **RAG pipeline** using **static code analysis**, symbol extraction, **source-code indexing**, and hybrid file-ranking techniques to identify issue-relevant files.
- Implemented an end-to-end **LLM orchestration workflow** with **automated program repair**, unified diff generation, rollback support, and `go vet/go test` validation for reliable software engineering automation.

POSITIONS OF RESPONSIBILITY

Marketing Head — Technex'25

Dec '24 – March '25

- Led **60+ member team**, driving initiatives that supported institute-level technical activities and outreach.
- Organized high-impact **corporate events** (LTTS, SBI, NHA, Tredence, AMD) with 2000+ student footfall.
- Steered Technex'25 finances (**Budget**, **Cash flow**) to optimize resource use and secured record sponsorship.

Marketing Executive — Technex'24

Jan '24 – March'24

- Engaged with marketing managers from various companies to secure sponsorships for Technex'24.
- Successfully raised local funds for Technex by pitching and negotiating with local businesses and organizations

HONOURS AND ACHIEVEMENTS

- Winner – Decision Science Track, Amex Campus Challenge'25** ranked **5th** nationally, earned PPI.
- Runner-up – MLWare, Technex'24** machine learning competition on real-world problem solving.
- JEE ADVANCED 2022** - secured **All India Rank 5809** among 1,50,000 shortlisted candidates.